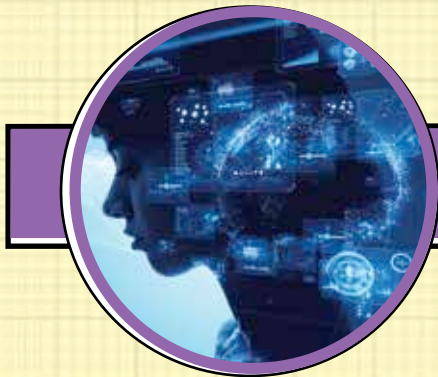


than each of these professionals focusing on only implementing the skills that they have, at a particular task. There are task tracking software built for this very purpose. There are a number of modern applications that allow people to collaboratively track the progress of tasks. Many professional software such as Adobe's Creative Cloud Suite, Tableau, Unity and Tally have evolved over the years to have collaboration features built in. Knowing how to use the collaborative aspects of these software can not only improve the final result, but also reduce back and forth. For larger companies, there are applications that mimic social networks, and can allow for

professional interactions within a closed environment. These kinds of applications provide a platform for employees of an organisation to interact with each other in an informal way. In fact social media companies have also realised that they can get in on the action, which was demonstrated by the roll out of Workplace, a collaborative environment for companies, by Facebook. A range of communication options, from instant messages, to video conferencing are offered by Workplace. Workplace has a free tier and a paid subscription, with a fees per user per month, which just offers better support and security. Apart from the apps mentioned below for collaboration, there are some tabletop games that are designed to improve teamwork, one that we would recommend is called The Mind.



COMPUTATIONAL THINKING

Computational thinking involves breaking down a problem into tiny bits, and finding out a repeatable path for solving them. Essentially, it involves designing algorithms to perform tasks, and is a skill that allows you to represent a process at its most granular level. Once this has been done, the actual task can be delegated to a computer, freeing up time for the user to engage in other tasks or activities. One of the biggest benefits of computational thinking is the ability to automate your work, and maybe even build the AI that performs your tasks. Don't worry, you will not go obsolete, as the AI will require monitoring and maintenance to perform as expected, at least at current technological levels. Now there are a number of applications that can enhance computational thinking abilities, and many of them are apps that teach you how to code. The important thing here is not learning the language, or acquiring deep magic skills, but the process of breaking down a task in such a manner that a computer can programmatically execute it. There are a number of steps required for computational thinking, and these are independent of the particular coding language used. First, the task has to be broken down into small pieces, or steps. This is known as problem decomposition. Then, it is necessary to remove unwanted details from the steps, a process known as abstraction. In the real world, the problems we encounter and the objects that we interact with are extremely complex and full of details. Computers do not need most of this information to execute a task. For example, if training a computer to solve a jigsaw puzzle, the computer only needs to know

the shape of each piece, and the pattern on it. The computer does not need to know the texture or the material used to make. In fact, depending on the type of jigsaw, a computer may be able to solve the puzzle without even knowing the pattern. This is abstraction. Another example is when teaching a computer to draw a cat, you only need to teach it the shape of the cat, and features such as eyes, ears, the tail and fur. The computer does not need to know that the cat makes a "meow" sound, or should not drink milk as an adult. Algorithmic thinking involves making a workflow, or a step by step approach for solving a problem. Then comes conditional logic, which is intimately tied to algorithmising a task. This means that a particular step or series of steps are performed if certain conditions are met. For example, it might be necessary to empty the garbage bin only when it is full. It would be a futile task to empty an empty garbage bin, and a waste of energy to empty a half full one. Then comes iterative thinking, which involves making multiple passes over a task to ensure that it is performed as expected. For example, a find and replace operation in an email can have unexpected results. It is necessary to check for these unexpected changes before sending the email out. Finally, there is debugging and testing, which is to check and ensure that the algorithm works as intended, even in edge cases. If there are any problems

in the process, they can be fixed during the debugging stage. Computational thinking done right can simplify your personal life as well. For example, a combination of NFC tags, smarthome devices and smartphones can be used to set up your home to automatically turn on the AC when someone leaves from work, set the temperature to a particular level when particular people are at home, dim the lights automatically at preset hours, and set the alarm for different times on particular days of the week.

Learn to algorithmise with these apps

- Mimo
- SoloLearn
- Pixel
Starships
- Shortcuts
- IFTTT
- Cargo Bot
- Knowin